

"lv"-able

A string is "lv"-able if it contains the contiguous substring "lv".

You are given a string s with N characters, and you want to make it "lv"-able in as few operations as possible.

You are allowed to do any of these operations:

- Remove any character at any position.
- Insert any character at any position.
- Replace any character by any other character at any position.
- Choose any consecutive interval of characters, and reverse the order of the characters in it.

Now make the string "lv"-able!

Input

The first line of input contains an integer N ($1 \leq N \leq 5 \cdot 10^5$), the number of characters in the initial string.

The second line contains the string s , which consists of N lowercase letters a - z.

Output

Print an integer: the minimum number of operations such that the string s becomes "lv"-able.

Scoring

Your solution will be tested on a set of test groups, each worth a number of points. Each test group contains a set of test cases. To get the points for a test group you need to solve all test cases in the test group.

Group	Points	Constraints
1	40	$N \leq 100$

2	60	No additional constraints.
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Explanation of Samples

In sample 1, you can reverse the substring “ov”, resulting in lvoable, which then contains “lv”.

In sample 2, we can replace the “e” with a “v”, which then contains “lv”.

In sample 3, the string already contains “lv”.

Sample Input 1

```
7
lovable
```



Sample Output 1

```
1
```



Sample Input 2

```
6
google
```



Sample Output 2

```
1
```



Sample Input 3

```
6
lvable
```



Sample Output 3

```
0
```